

× PROOF-OF-REASONING

● MAINNET · LIVE

ERC-8004 REFERENCE IMPL

TuringVault

AI-powered RWA portfolio application on Mantle —
every proposed allocation **survives adversarial
challenge** before execution, anchored on chain.

ASSET CLASS

Tokenized Treasuries

USDT0 · mETH · USDY

CHAIN

Mantle Mainnet

6 contracts deployed · 5/6
Sourcify-verified `perfect` (Router
source drifted post-deploy)

VERIFICATION

3-model + ERC-8004

GLM-5 · Claude 4.6 · Gemini 3.5

WHAT MANTLE JUDGES SAID MATTERS

The hackathon brief lists **three defining features**. We built our entire stack around them.

"On-chain benchmarking of AI · ERC-8004 agent identity standard · radical transparency" — Mantle Turing Test 2026 brief

01 On-chain benchmarking of AI

Every cycle writes 4 attestation TXs to Mantle Mainnet (proposal · validation · decisionLog · reputation). 840+ cycles logged on-chain (700+ settled with scored outcomes), each with full reasoning hashes. The scheduled cron's public GitHub Actions workflow log gives judges a verifiable artefact, not a claim.

02 ERC-8004 agent identity standard

Reference implementation deployed on Mantle Mainnet — non-transferable identity NFT with auto-updating tokenURI, on-chain reputation accumulating from settled outcomes, validation registry for pre-action attestation. Ready to interop with Mantle's official Agent ID issuance.

03 Radical transparency

Public scheduled cron · IPFS-pinned reasoning · live mascot reflecting real cycle freshness · /challenge page lets visitors probe the agent's gates · /discipline page shows post-execution proof per cycle · honesty-rule documented (.kiro/steering/no-lying-about-state.md) and enforced.

Live-stream ready: GitHub Actions workflow runs on a public schedule (best-effort hourly), judges can drop into any past or current run.

WHY THIS MATTERS

AI agents allocate real capital. Nobody can verify why.

TODAY'S AUTONOMOUS AGENT (SCHEDULED CRON)

- Receives capital
- Black-box LLM produces decision
- Signs and broadcasts on-chain
- Reports PnL screenshot

No audit trail · No challenge · No accountability.

TURINGVAULT

- Receives capital
- **Three independent models challenge each other**
- **Every decision hashed → IPFS → Mantle**
- **Post-execution proof verification (Discipline Layer)**

Refusal-with-proof > profitable execution without one.

For RWA allocation specifically — DAO treasuries, DeFi-native funds, compliance-conscious operators — accountability is not a feature. It's the entire reason RWA assets exist.

OUR POSITIONING

Not a trading bot. An accountable RWA portfolio manager.

RISK-ON LEG

mETH

Mantle native ETH staking — real yield

TREASURY ALLOCATION

USDT0

LayerZero-bridged Tether — Treasury-collateralised

RISK-OFF FLOOR

USDT

Stablecoin · pre-allocation buffer

TRADE WIN RATE

68%

Executed swaps only · 62 / 91 (excludes holds & unexecuted intents)

RWA EXPOSURE

65% NAV

Tokenized Treasuries (USDT0) · live wallet

DECISION QUALITY

49.5%

Correct trades + correct risk-off holds / realized outcomes

Compliance-conscious DAOs need an agent that can *prove its restraint*, not just its profits. Every blocked allocation has on-chain reasoning anchored to Mantle and pinned to IPFS.

HOW A DECISION FLOWS

Every proposed allocation passes **three independent models** and **three on-chain attestations**. Then a fourth gate verifies the execution itself.

1. ANALYST

Z.ai GLM-5

Proposes allocation from unified market context (CoinGecko, Nansen MCP, Byreal Perps funding/OI, DeFiLlama TVL, Elfa social attention).

2. VALIDATOR

Anthropic Claude 4.6

Adversarial gate. Default REJECT. Burden of proof on the proposer. R/R $\geq 1.5:1$ mandatory.

3. ARBITER

Google Gemini 3.5

Tiebreaker for soft confidence disputes. Hard validator veto is final.

```

GitHub Actions cron (scheduled)
(public verifiable log) | ▼
5 data sources: CoinGecko · Nansen
MCP · Byreal Perps · DeFiLlama ·
Elfa social attention
| ▼
Signal Engine: 5 structured signals
(funding · on-chain flow · yield spread · liq map · social) | ▼
unified market context → Analyst → Validator → Arbiter | ▼
4 on-chain attestations:
proposal · validation · decisionLog · reputation | ▼
RWA Allocator (LLM-reviewed ·
deterministic idle-parking controls) | ▼ Merchant Moe LB v2.2 swap →
Discipline Layer 3-gate post-exec
verification

```

VERIFIABLE ON-CHAIN RIGHT NOW

Everything below is verifiable on Mantlescan.

700

SETTLED OUTCOMES

68%

TRADE WIN RATE · EXECUTED

65%

NAV IN RWA

49.5%

DECISION QUALITY

AGENT PIPELINE · LIFETIME (STAGE COUNTS)

700 decisions ▶ 152 consensus-approved (22%) ▶ 91 executed ▶ 62 won (68%)

548 blocked · 471 deterministic risk gates · 41 adversarial validator · 36 other — selective by design: capital committed only on full 3-agent consensus.

Snapshot 2026-06-27. Live values refresh on the dashboard (frontend-seven-beta-46.vercel.app · /api/performance). Trade win rate counts executed directional swaps only (62/91); decision quality credits correct risk-off holds. NAV % drifts cycle to cycle as the agent rebalances; settled-outcome count increases monotonically.

SMART CONTRACTS (SOURCIFY)

Identity	0x6f86...28bD
DecisionLog	0x7bCd...cfbB5
ValidationRegistry	0x6841...63b6

PUBLIC ARTIFACTS

First RWA swap (Mantlescan)	0x0af2336...3e09de
Scheduled cron (GitHub Actions)	
Agent Gnosis workflow	public_log

RADICAL TRANSPARENCY

We hold ourselves **to the same standard** we ask of the agent.

A project that promises Proof-of-Reasoning cannot afford fabricated metrics. Every claim on the dashboard is traceable to a contract read, a settled outcome, or a public file.

LIVE	AI consensus pipeline	840+ cycles · live
LIVE	ERC-8004 identity + reputation	5/6 Sourcify-verified
LIVE	Scheduled cron via GitHub Actions	public workflow log
LIVE	Discipline Layer post-execution gate	3 gates fire each cycle
LIVE	RWA allocation to USDT0	first swap 0x0af2336...

Source-of-truth: `.kiro/steering/no-lying-about-state.md`

Application layer for **DAO-grade RWA allocation**, built natively on Mantle's primitives.

AI & RWA Path B scoring rewards clear asset scope, coherent users, end-to-end UX, and auditable AI outputs. TuringVault ships that as a live Mantle application, not a chatbot wrapper.

METH

Native ETH staking

Risk-on RWA leg with real on-chain yield, sourced from Mantle's own LST

USDT0

LayerZero omnichain Tether

\$1.1M Merchant Moe LB v2.2 pool (as of May 2026) — only liquid Treasury-collateralised target on Mantle

CRON

GitHub Actions scheduled

~0.08 MNT / full cycle gas (8 TXs incl. 3 swaps) ≈ \ \$0.05 · free-tier sustainable, verifiable in Mantlescan

SOURCIFY

5/6 Sourcify-verified `perfect`

Judge-readable source on Mantle Explorer (Router pending)

INNOVATION IN TECHNICAL APPROACH

- Three-model adversarial consensus with cross-provider diversity (Bedrock + Vertex AI)
- ERC-8004 reference implementation — auto-updating tokenURI per cycle
- Synrail-inspired Discipline Layer — TX proof + freshness + drift gates after every swap
- RWA Allocator: LLM-reviewed allocation + deterministic idle parking controls

WHO BUILT THIS AND WHAT'S NEXT

A solo developer. A repeatable pattern.

TEAM

USBVadik

Solo developer · ex-Synrail (autonomous-agent verification framework, also OSS)

github.com/USBVadik/TuringVault-Core

github.com/USBVadik/synrail

x.com/a_seven_life

STACK DISCIPLINE

→ Spec-first workflow (9 specs, >340 acceptance criteria)

→ 177/177 unit tests · CI on every push

→ Honesty rule enforced as steering doc

→ Every deploy verifiable on Mantlescan + GitHub Actions log

ROADMAP (POST-HACKATHON)

Q3 2026

Vault contract for public deposits · DAO treasury onboarding · multi-asset RWA basket

Q4 2026

Cross-agent reputation marketplace · ERC-8004 identity graph · per-decision IPFS streaming

2027

Open the Discipline Layer as an SDK · third-party agents stake reputation through TuringVault

LIVE DEMO

frontend-seven-beta-46.vercel.app

SOURCE

github.com/USBVadik/TuringVault-Core

CRON LOG

[Actions](#) / [Agent Cycle](#)

"The most valuable thing an AI RWA portfolio manager can do is **prove why it didn't allocate.**"